

Your grade: 90%

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[Next item →](#)

1. Using the notation for mini-batch gradient descent. To what of the following does $a^{[2]\{4\}(3)}$ correspond?

1 / 1 point

- ☐ The activation of the fourth layer when the input is the second example of the third mini-batch.
- ☐ The activation of the third layer when the input is the fourth example of the second mini-batch.
- ☐ The activation of the second layer when the input is the fourth example of the third mini-batch.
- ☒ The activation of the second layer when the input is the third example of the fourth mini-batch.

✔ Nice work

Yes. In general $a^{[l]\{t\}(k)}$ denotes the activation of the layer l when the input is the example k from the mini-batch t .

2. Which of these statements about mini-batch gradient descent do you agree with?

1 / 1 point

- ☐ You should implement mini-batch gradient descent without an explicit for-loop over different mini-batches, so that the algorithm processes all mini-batches at the same time (vectorization).
- ☐ Training one epoch (one pass through the training set) using mini-batch gradient descent is faster than training one epoch using batch gradient descent.
- ☒ One iteration of mini-batch gradient descent (computing on a single mini-batch) is faster than one iteration of batch gradient descent.

✔ Nice work

3. We usually choose a mini-batch size greater than 1 and less than m , because that way we make use of vectorization but not fall into the slower case of batch gradient descent.

0 / 1 point

- ☒ One iteration of mini-batch gradient descent (computing on a single mini-batch) is faster than one iteration of batch gradient descent.

✓ Nice work

3. We usually choose a mini-batch size greater than 1 and less than m , because that way we make use of vectorization but not fall into the slower case of batch gradient descent.

0 / 1 point

☒ False

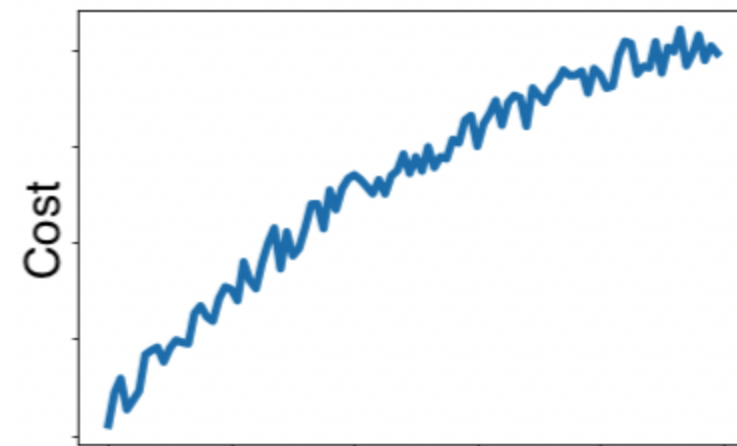
☐ True

✗ Try again

Incorrect. Precisely by choosing a batch size greater than one we can use vectorization; but we choose a value less than m so we won't end up using batch gradient descent.

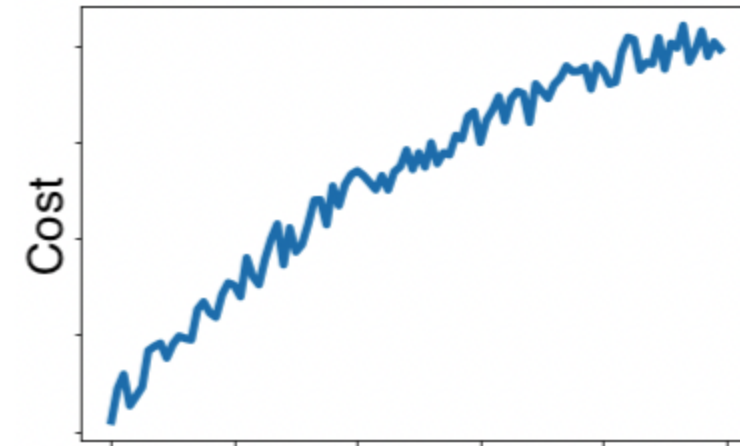
4. While using mini-batch gradient descent with a batch size larger than 1 but less than m the plot of the cost function J looks like this:

1 / 1 point



4. While using mini-batch gradient descent with a batch size larger than 1 but less than m the plot of the cost function J looks like this:

1 / 1 point



Which of the following do you agree with?

- ☒ No matter if using mini-batch gradient descent or batch gradient descent something is wrong.
- ☐ If you are using mini-batch gradient descent or batch gradient descent this looks acceptable.
- ☐ If you are using batch gradient descent, this looks acceptable. But if you're using mini-batch gradient descent, something is wrong.
- ☐ If you are using mini-batch gradient descent, this looks acceptable. But if you're using batch gradient descent, something is wrong.

✓ Nice work

Yes. The cost is larger than when the process started, this is not right at all.

5. Suppose the temperature in Casablanca over the first two days of March are the following:

1 / 1 point

- ☐ If you are using mini-batch gradient descent, this looks acceptable. But if you're using batch gradient descent, something is wrong.

✔ Nice work

Yes. The cost is larger than when the process started, this is not right at all.

5. Suppose the temperature in Casablanca over the first two days of March are the following:

1 / 1 point

March 1st: $\theta_1 = 30^\circ \text{ C}$

March 2nd: $\theta_2 = 15^\circ \text{ C}$

Say you use an exponentially weighted average with $\beta = 0.5$ to track the temperature: $v_0 = 0$, $v_t = \beta v_{t-1} + (1 - \beta) \theta_t$. If v_2 is the value computed after day 2 without bias correction, and $v_2^{\text{corrected}}$ is the value you compute with bias correction. What are these values?

- ☐ $v_2 = 20, v_2^{\text{corrected}} = 15$.
- ☒ $v_2 = 15, v_2^{\text{corrected}} = 20$.
- ☐ $v_2 = 15, v_2^{\text{corrected}} = 15$.
- ☐ $v_2 = 20, v_2^{\text{corrected}} = 20$.

✔ Nice work

Correct. $v_2 = \beta v_{t-1} + (1 - \beta) \theta_t$ thus $v_1 = 15, v_2 = 15$. Using the bias correction $\frac{v_t}{1 - \beta^t}$ we get $\frac{15}{1 - (0.5)^2} = 20$.

6. Which of the following is true about learning rate decay?

1 / 1 point

- ☐ We use it to increase the size of the steps taken in each mini batch iteration.

- ☐ $v_2 = 15, v_2^{\text{corrected}} = 15.$
☐ $v_2 = 20, v_2^{\text{corrected}} = 20.$

✔ Nice work

Correct. $v_2 = \beta v_{t-1} + (1 - \beta) \theta_t$ thus $v_1 = 15, v_2 = 15$. Using the bias correction $\frac{v_t}{1 - \beta^t}$ we get $\frac{15}{1 - (0.5)^2} = 20$.

6. Which of the following is true about learning rate decay?

1 / 1 point

- ☐ We use it to increase the size of the steps taken in each mini-batch iteration.
☒ The intuition behind it is that for later epochs our parameters are closer to a minimum thus it is more convenient to take smaller steps to prevent large oscillations.
☐ It helps to reduce the variance of a model.
☐ The intuition behind it is that for later epochs our parameters are closer to a minimum thus it is more convenient to take larger steps to accelerate the convergence.

✔ Nice work

Correct. Reducing the learning rate with time reduces the oscillation around a minimum.

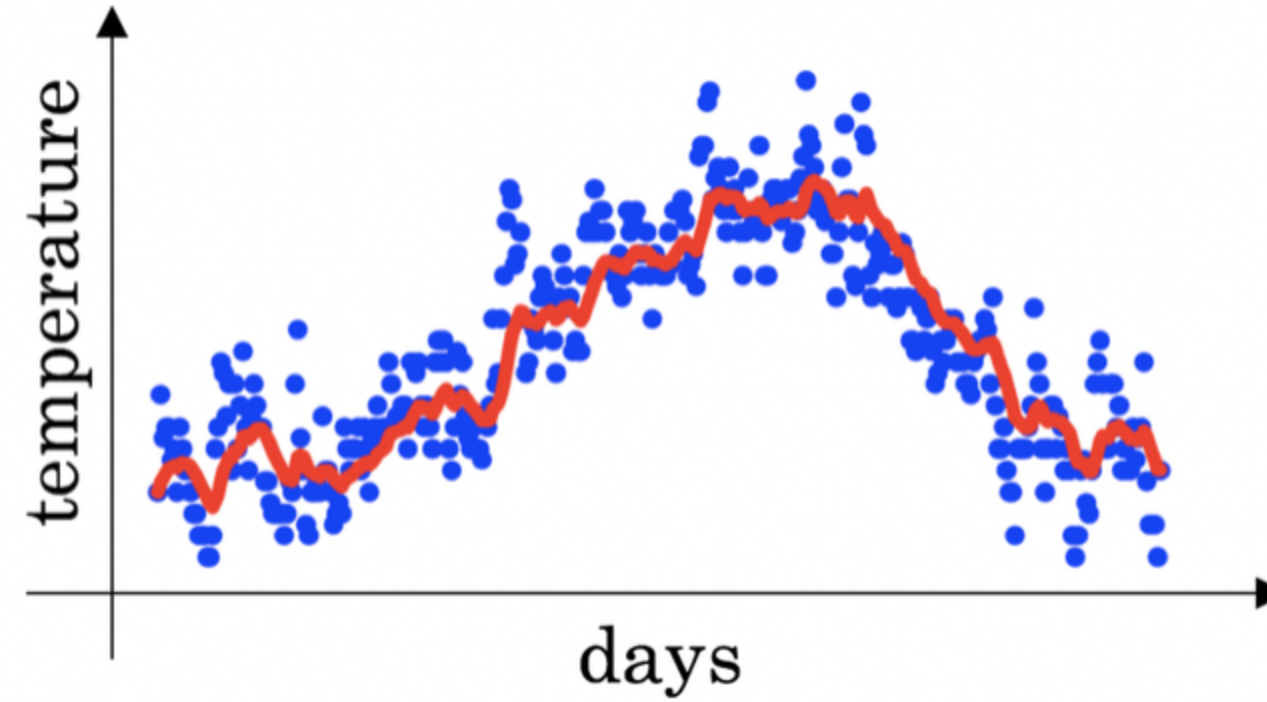
7. You use an exponentially weighted average on the London temperature dataset. You use the following to track the temperature: $v_t = \beta v_{t-1} + (1 - \beta) \theta_t$. The red line below was computed using $\beta = 0.9$. What would happen to your red curve as you vary β ? (Check the two that apply)

1 / 1 point



7. You use an exponentially weighted average on the London temperature dataset. You use the following to track the temperature: $v_t = \beta v_{t-1} + (1 - \beta)\theta_t$. The red line below was computed using $\beta = 0.9$. What would happen to your red curve as you vary β ? (Check the two that apply)

1 / 1 point



- ☐ Decreasing β will shift the red line slightly to the right.
- ☒ Increasing β will shift the red line slightly to the right.

✓ Nice work

True, remember that the red line corresponds to $\beta = 0.9$. In the lecture we had a green line $\beta = 0.98$ that is slightly shifted to the right.

- ☒ Decreasing β will create more oscillation within the red line.

✓ Nice work

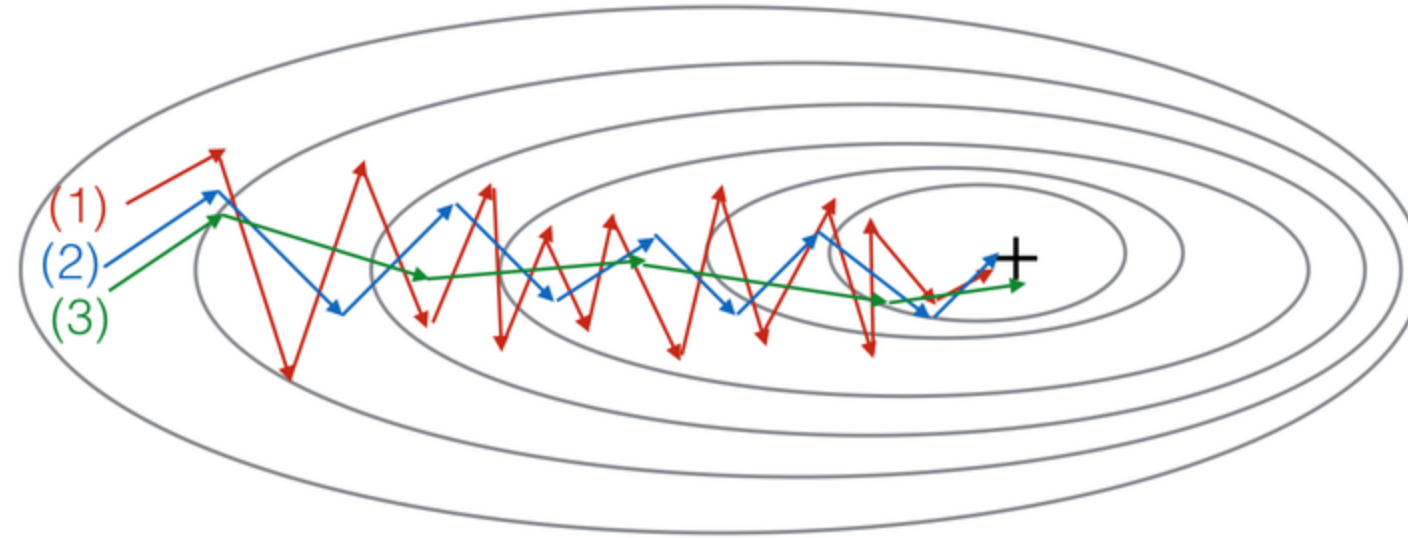
True, remember that the red line corresponds to $\beta = 0.9$. In lecture we had a yellow line $\beta = 0.98$ that had a lot of oscillations.

- ☐ Increasing β will create more oscillations within the red line.

☐ Increasing β will create more oscillations within the red line.

8. Consider this figure:

1 / 1 point



These plots were generated with gradient descent; with gradient descent with momentum ($\beta = 0.5$); and gradient descent with momentum ($\beta = 0.9$). Which curve corresponds to which algorithm?

- ☐ (1) is gradient descent. (2) is gradient descent with momentum (large β). (3) is gradient descent with momentum (small β)
- ☐ (1) is gradient descent with momentum (small β). (2) is gradient descent. (3) is gradient descent with momentum (large β)
- ☒ (1) is gradient descent. (2) is gradient descent with momentum (small β). (3) is gradient descent with momentum (large β)
- ☐ (1) is gradient descent with momentum (small β), (2) is gradient descent with momentum (small β), (3) is gradient descent

✓ Nice work

9. Suppose batch gradient descent in a deep network is taking excessively long to find a value of the

1 / 1 point

9. Suppose batch gradient descent in a deep network is taking excessively long to find a value of the parameters that achieves a small value for the cost function $\mathcal{J}(W^{[1]}, b^{[1]}, \dots, W^{[L]}, b^{[L]})$. Which of the following techniques could help find parameter values that attain a small value for $\mathcal{J}$? (Check all that apply)

1 / 1 point

- ☒ Try better random initialization for the weights

✔ Nice work

Yes. As seen in previous lectures this can help the gradient descent process to prevent vanishing gradients.

- ☒ Try using gradient descent with momentum.

✔ Nice work

Yes. The use of momentum can improve the speed of the training. Although other methods might give better results, such as Adam.

- ☒ Normalize the input data.

✔ Nice work

Yes. In some cases, if the scale of the features is very different, normalizing the input data will speed up the training process.

- ☐ Add more data to the training set.

10. Which of the following are true about Adam?

1 / 1 point

- ☒ Adam combines the advantages of RMSProp and momentum.
- ☐ The most important hyperparameter on Adam is ϵ and should be carefully tuned.
- ☐ Adam can only be used with batch gradient descent and not with mini-batch gradient descent.
- ☐ Adam automatically tunes the hyperparameter α .

✔ Nice work

True. Precisely Adam combines the features of RMSProp and momentum that is why we use two-parameter β_1 and β_2 , besides ϵ .